

PROBABILITY & STATISTICS

- **Expectation:** - Expectation = $E(X) = \sum_{\text{all } x} xf(x) = \int xf(x)$

- (1) $E(c) = c$, c is a constant.
- (2) $E(cX) = cE(X)$, c is a constant.
- (3) $E(b + cX) = b + cE(X)$, b, c are constants.
- (4) $E(X + Y) = E(X) + E(Y)$.

- **Variance:** - Variance = $\pm^2 = E((X - \mu)^2) = E(X^2) - (E(X))^2$.

- (1) $\text{Var}(c) = 0$, c is a constant.
- (2) $\text{Var}(b + cX) = c^2\text{Var}(X)$, b, c are constants.
- (3) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

- **Moment Generating Function:** -

Moment Generating Function = $M_X(t) = E(e^{tX}) = \sum_{\text{all } x} e^{tx} f(x)$.

If $M_X(t)$ is the moment generating function then:

- (1) $E(X) = M'_X(0)$
- (2) $E(X^2) = M''_X(0)$
- (3) Variance = $\pm^2 = E(X^2) - (E(X))^2 = M''_X(0) - (M'_X(0))^2$.

<u>Distribution</u>	<u>Density</u>	<u>M.G.F</u>	<u>Mean</u>	<u>Variance</u>
<i>Geometric</i>	$f(x) = (1-p)^x p$	$M_x(t) = \frac{pe^t}{1-(1-p)e^t}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
<i>Binomial</i>	$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$	$M_x(t) = (1-p + pe^t)^n$	np	$np(1-p)$
<i>Negative Binomial</i>	$f(x) = \binom{n-1}{x-1} p^x (1-p)^{n-x}$	$M_x(t) = \frac{(pe^t)^n}{(1-p + pe^t)^n}$	$\frac{n}{p}$	$\frac{n(1-p)}{p^2}$
<i>Hypergeometric</i>	$f(x) = \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}}$		$\frac{nr}{N}$	$\frac{nr(N-r)(N-n)}{N^2(N-1)}$
<i>Poisson</i>	$f(x) = \frac{e^{-a} a^x}{x!}$	$M_x(t) = e^{a(e^t-1)}$	a	a
<i>Bernoulli</i>	$f(x) = p^x (1-p)^{1-x}$	$M_x(t) = 1-p + pe^t$	p	$p(1-p)$
